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How Monetary Policy Shaped the Housing Boom

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1 Executive summary

The paper asks a sharp question: *why did the U.S. housing boom continue, and private-label securitization (PLS) surge, during the 2004–2006 monetary tightening cycle?* The authors argue that an important part of the answer is a **composition channel** operating through banks’ **core deposits**.

- A monetary tightening raises the short rate. Because deposits provide liquidity services and deposit pricing is imperfectly competitive, banks do *not* raise deposit rates one-for-one. As a result, **deposit spreads** (market rate minus deposit rate) increase.
- Higher deposit spreads make deposits less attractive to households and firms, causing **deposit outflows** and weaker deposit growth—especially at banks with **higher exposure** to this “deposits channel”.
- Because core deposits are a key input into bank balance-sheet mortgage lending (especially long-duration real estate lending), weaker deposit growth leads to **lower bank portfolio mortgage lending**.
- The mortgage market does not necessarily shrink one-for-one. Instead, credit supply shifts toward **PLS and nonbank origination**, which are less constrained by deposit funding.
- Therefore, monetary tightening can have a limited effect on *total* mortgage origination while substantially changing the *composition* of mortgage finance toward more fragile forms.

Crucially, the paper does not rely on time-series correlations alone. It identifies the mechanism in the **cross section** using *pre-boom* measures of how sensitive local deposit spreads are to the policy rate (“deposit-spread betas”). The core empirical result is:

Counties whose mortgage markets were served by high-beta banks experienced (i) weaker bank portfolio mortgage growth and (ii) a larger shift toward PLS and non-bank lending during 2003–2006.

2 Institutional background: the mortgage-finance objects in the paper

A recurring theme is that “mortgage lending” is not a single object. The paper separates originations into economically distinct channels.

2.1 Mortgage origination categories

Using HMDA originations and additional sources, the paper distinguishes:

1. **Bank portfolio lending**: the originating bank keeps the mortgage on its balance sheet.
2. **GSE lending**: mortgages securitized/guaranteed by government-sponsored enterprises (GSEs).

3. **Private-label securitization (PLS)**: mortgages securitized without GSE guarantees. PLS can be originated by banks or by nonbanks.
4. **Nonbank lending**: originations by lenders that do not fund themselves primarily with insured deposits.

2.2 Why deposits matter for bank mortgage lending

Banks' *core deposits* are typically:

- **cheap** relative to market funding because depositors accept a lower yield in exchange for liquidity services, and
- **interest-rate-insensitive** relative to wholesale funding (deposit rates move sluggishly).

These features make deposits a natural funding source for long-duration assets such as mortgages. A key implication is that shocks to deposit supply can translate into shocks to bank mortgage credit supply.

3 Mechanism: deposit spreads, betas, and the deposits channel

3.1 Deposit spreads and pass-through

Let the (short) market rate be the Fed funds rate, denoted by FF_t . Let $r_{b,t}^D$ be the interest rate paid by branch b on a given deposit product. Define the **deposit spread** as

$$\text{Spread}_{b,t} \equiv FF_t - r_{b,t}^D. \quad (1)$$

A larger spread means depositors forgo more interest relative to the market rate, i.e., deposits are “more expensive” for depositors and “cheaper” for banks.

Suppose empirically that deposit spreads respond to policy changes via

$$\Delta \text{Spread}_{b,t} \approx \beta^{\text{spread}} \Delta FF_t.$$

Then the implied pass-through of the policy rate into the deposit rate is

$$\Delta r_{b,t}^D = \Delta FF_t - \Delta \text{Spread}_{b,t} \approx (1 - \beta^{\text{spread}}) \Delta FF_t. \quad (2)$$

Hence, a **higher spread beta** β^{spread} means **lower pass-through** to deposit rates and therefore a stronger increase in deposit spreads when the Fed tightens.

3.2 Predictions (what should move with beta?)

Holding mortgage *demand* fixed, the deposits channel implies:

1. **First stage**: when FF_t rises, deposit spreads rise more in high-beta places/banks.

2. **Deposit supply:** deposit growth is lower where spreads rise more (high beta).
3. **Bank lending:** bank portfolio mortgage growth is lower where deposits contract more.
4. **Market composition:** PLS and nonbank shares rise in high-beta places as the mortgage market substitutes away from deposit-funded balance-sheet lending.

The paper tests these predictions sequentially and at multiple aggregation levels (branch $\rightarrow$ bank $\rightarrow$ county).

4 Data and measurement

4.1 Deposit rates and branch-level deposits

- **Branch deposit rates:** RateWatch data on deposit interest rates for checking, savings, and small time deposits.
- **Branch deposits (quantities):** FDIC Summary of Deposits (SoD), used to measure branch deposit growth.

4.2 Bank balance sheets and funding

- **Call Reports:** bank-level balance-sheet items (core deposits, loans by category, securities, cash, time deposits, wholesale funding).
- The paper constructs a **bank interest-expense spread:** FF_t minus the ratio of annualized interest expense to average assets.

4.3 Mortgage originations and market shares

- **HMDA:** county-level mortgage originations split into bank portfolio, bank PLS, nonbank PLS, etc.; most tests focus on **non-GSE** lending.
- **House prices:** Zillow house-price indices (used for robustness and pre-trend controls).
- **Policy stance for motivation:** one-year Treasury forward rate (Fig. 1).

5 All models in the paper (equations and econometric specifications)

This section lists the paper’s core empirical models and explains the logic behind each one. Notation is kept consistent across levels of aggregation.

5.1 (1) Estimating branch deposit-spread betas (pre-boom)

For each county c and branch b in that county, and separately for each deposit product, the paper estimates:

$$\Delta\text{Spread}_{b,c,t} = \alpha_c + \sum_{\tau=0}^3 \beta_{c,\tau} \Delta\text{FF}_{t-\tau} + \varepsilon_{b,c,t}. \quad (1)$$

- $\Delta\text{Spread}_{b,c,t}$ is the change in the branch's deposit spread from $t - 1$ to t .
- Three lags allow spreads to adjust over a year.
- The regression is run on pre-boom data (1997–2002) using all branches in county c . The estimated coefficients are county-level; each branch in that county inherits the same county-product beta.
- The paper sums $\beta_{c,\tau}$ over τ to obtain a single beta for each product, winsorizes to reduce outliers due to estimation error, and then averages across products to get a single **branch beta** BranchBeta_b .

5.2 (2) Estimating bank interest-expense spread betas (pre-boom)

Using Call Reports for bank j over 1986–2002, the paper estimates:

$$\Delta\text{Spread}_{j,t} = \alpha_j + \sum_{\tau=0}^3 \beta_{j,\tau} \Delta\text{FF}_{t-\tau} + \varepsilon_{j,t}, \quad (2)$$

where $\text{Spread}_{j,t}$ is the interest-expense spread. Summing $\beta_{j,\tau}$ produces:

$$\text{BankBeta}_j \equiv \sum_{\tau=0}^3 \beta_{j,\tau}.$$

Interpretation: high BankBeta_j means that when policy tightens, the bank's funding cost (interest expense) rises *less* relative to the market rate, so the spread increases *more*. This stronger rise in spreads is associated with stronger deposit outflows in the tightening cycle.

5.3 (3) Constructing county exposure to the deposits channel

County exposure is a weighted average of bank betas using base-year (2002) portfolio mortgage lending shares:

$$\text{CountyBeta}_c = \sum_j s_{j,c} \text{BankBeta}_j, \quad (3)$$

where $s_{j,c}$ is bank j 's 2002 share of portfolio mortgage lending in county c . **Key point:** exposure is measured *before* the boom, reducing concerns that betas are chosen in response to the boom.

5.4 (4) Branch-level “first stage”: beta predicts deposit-spread changes

During 2003–2006, for branch b :

$$\Delta \text{DepositSpread}_b = \alpha + \gamma \text{BranchBeta}_b + \varepsilon_b. \quad (4)$$

Goal: show that pre-period betas predict realized spread changes out of sample.

5.5 (5) Branch-level deposit supply: beta predicts deposit growth

$$\text{DepositGrowth}_b = \alpha + \gamma \text{BranchBeta}_b + \varepsilon_b, \quad (5)$$

with deposit growth measured from mid-2003 to mid-2007. A key robustness adds **bank fixed effects**, comparing different branches of the *same* bank across counties.

5.6 (6) Bank-level balance-sheet responses (reduced form)

For a generic balance-sheet component Y_j :

$$\frac{\Delta Y_j}{\text{CoreDep}_j} = \alpha + \gamma \text{BankBeta}_j + \delta X_j + \varepsilon_j, \quad (6)$$

with changes from 2003 to 2006 and the denominator equal to initial core deposits.

5.7 Bank-level 2SLS pass-through (conceptual model)

Table 3 also reports a 2SLS design that uses BankBeta_j as an instrument for core-deposit growth:

$$\text{First stage: } \frac{\Delta \text{CoreDep}_j}{\text{CoreDep}_j} = \pi_0 + \pi_1 \text{BankBeta}_j + \pi_2 X_j + u_j, \quad (3)$$

$$\text{Second stage: } \frac{\Delta Y_j}{\text{CoreDep}_j} = \alpha + \lambda \left(\frac{\widehat{\Delta \text{CoreDep}_j}}{\text{CoreDep}_j} \right) + \delta X_j + \varepsilon_j. \quad (4)$$

λ is the *deposit pass-through* parameter (dollars of Y per dollar of deposit growth induced by the instrument).

5.8 (7) County-level outcomes: lending composition and the shift to PLS

For county outcomes Y_c (portfolio lending growth, PLS share change, nonbank share change, etc.):

$$\Delta Y_c = \alpha + \gamma \text{CountyBeta}_c + \delta X_c + \varepsilon_c. \quad (7)$$

Controls are added in layers (market size, market structure, and demand/economic-growth controls).

6 Interpreting figures and tables

6.1 Fig. 1: Time-series motivation

Fig. 1 shows that the PLS share rises when the stance of policy (one-year forward rate) moves toward tightening. This is a motivation plot: it suggests that tightening and PLS growth coincide, but it does not by itself identify causality.

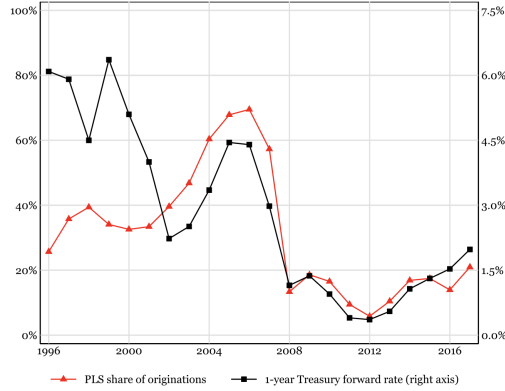


Fig. 1. Private-label securitization and monetary policy
The figure plots the private-label securitization (PLS) share of originations against the one-year Treasury forward rate, a measure of the stance of monetary policy. The PLS share is computed as the ratio of PLS originations divided by the sum of PLS originations and bank portfolio loans. GSE originations are excluded to focus on the shift from portfolio loans to PLS. PLS originations are from the Securities Industry and Financial Markets Association (SIFMA). Portfolio loans are from the Home Mortgage Disclosure Act (HMDA) dataset. The one-year forward rate is from the Federal Reserve. The sample is from 1996 to 2017.

Table 1

Summary statistics: branch, bank, and county characteristics.

This table provides summary statistics at the branch, bank, and county levels. All panels provide breakdowns by high and low beta partitioned by the median beta for the respective sample. Panel A presents branch-level summary statistics on branch beta and the change in deposit spread by product (in percentage points) from January 2003 to December 2006. The data are from Ratewatch. Panel B presents branch-level summary statistics on branch size (in millions of dollars) and deposit growth (in percent) from June 2003 to June 2007. The data are from the FDIC. Panel C presents bank-level summary statistics on bank beta and the change in deposit spread (in percentage points), the percentage growth in core deposits (calculated as dollar-difference changes from 2003 to 2006 divided by 2003 core deposits), real estate loans, commercial and industrial (C&I) loans, personal loans, cash, securities, large time deposits, and wholesale funding from January 2003 to December 2006. The data are quarterly from the U.S. Call Reports. Panel D presents county-level summary statistics on county beta and the growth in bank portfolio lending, total bank lending, total lending (in percent) and the change in the private-label market share and the bank market share (in percentage points) from 2003 to 2006. All lending measures exclude GSE lending except for "Δ Log Bank GSE lending" and "Δ Log Nonbank GSE lending." For all variables we conduct a T-test to determine whether the low and high beta samples are statistically significantly different from each other. We report the results in column (5) of each panel; *** and ** denote significance levels at 1% and 5%, respectively. The data are based on mortgage originations reported under HMDA. The table also reports county characteristics measured in 2002. The data are from the Bureau of Labor Statistics and the Census Bureau.

	All branches		Low beta	High beta	T-test
	Mean	St.Dev.	Mean	Mean	Sig. Level
	(1)	(2)	(3)	(4)	(5)
Panel A: Branch-level characteristics (rates)					
Branch beta	0.581	(0.077)	0.523	0.640	***
Δ Savings deposit spread	3.374	(0.992)	3.226	3.522	***
Δ Small time deposit spread	1.709	(0.664)	1.613	1.806	***
Δ Interest checking deposit spread	4.044	(0.386)	3.988	4.100	***
Observations	5,501		2,750	2,751	
Panel B: Branch-level characteristics (amounts)					
Branch beta	0.590	(0.088)	0.525	0.656	***
Deposit growth	0.234	(0.468)	0.262	0.205	***
Deposits (mill. \$; 2007)	76.325	(600.329)	66.673	85.938	***
Observations	58,546		29,215	29,331	
Panel C: Bank characteristics					
Bank beta	0.626	(0.094)	0.556	0.695	***
Δ Core deposit spread	3.236	(0.690)	3.015	3.457	***
Δ Core deposits/Core deposits	0.246	(0.314)	0.298	0.195	***
Δ Real estate loans/Core deposits	0.306	(0.333)	0.360	0.253	***
Δ C&I loans/Core deposits	0.049	(0.084)	0.058	0.040	***
Δ Personal loans/Core deposits	-0.001	(0.035)	-0.001	-0.001	***
Δ Cash/Core deposits	0.009	(0.086)	0.016	0.001	***
Δ Securities/Core deposits	0.050	(0.145)	0.066	0.034	***
Δ Large time/Core deposits	0.125	(0.151)	0.150	0.099	***
Δ Wholesale funding/Core deposits	0.040	(0.086)	0.045	0.034	***
Observations	6,463		3,231	3,232	
Panel D: County characteristics					
County beta	0.545	(0.056)	0.501	0.588	***
Δ Bank portfolio lending	0.126	(0.505)	0.155	0.098	***
Δ Bank lending	0.404	(0.426)	0.444	0.364	***
Δ Total lending	0.428	(0.365)	0.444	0.411	**
Δ PLS lending share	0.114	(0.144)	0.105	0.123	***
Δ Nonbank lending share	0.019	(0.129)	0.005	0.032	***
Log total lending (2002)	5.109	(1.871)	5.282	4.937	***
Log employment (2002)	9.209	(1.523)	9.349	9.069	***
Log median income (2002)	10.464	(0.241)	10.461	10.466	***
Log bank portfolio lending (2002)	4.356	(1.817)	4.473	4.239	***
PLS lending share (2002)	0.497	(0.173)	0.525	0.468	***
Nonbank lending share (2002)	0.266	(0.150)	0.295	0.236	***
County deposit-weighted beta (2002)	0.578	(0.069)	0.550	0.606	***
Top 4 lenders' share (2002)	0.505	(0.172)	0.479	0.531	***
Δ Bank GSE lending	-0.086	(0.515)	-0.095	-0.077	***
Δ Nonbank GSE lending	-0.146	(0.709)	-0.138	-0.153	***
Δ Employment	0.039	(0.082)	0.042	0.035	**
Δ Income	0.096	(0.045)	0.095	0.097	***
Observations	3,033		1,516	1,517	

6.2 Table 1: Summary statistics (what differs between low- and high-beta places?)

Table 1 summarizes key objects and splits samples at the median beta.

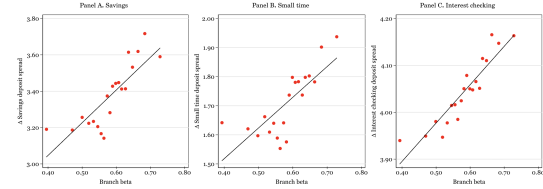
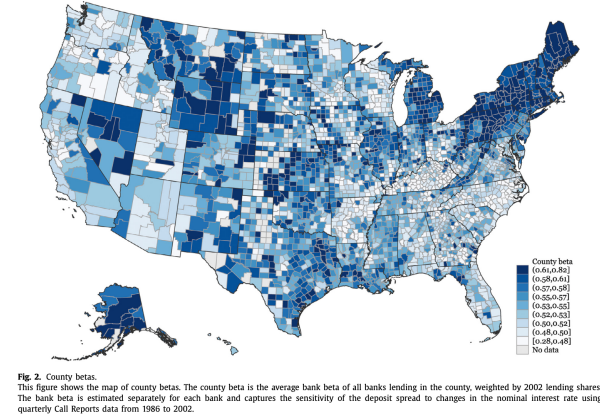
- **Branch rates (Panel A):** high-beta branches have higher betas (0.640 vs. 0.523) and larger spread increases from 2003–2006: savings spreads rise by 3.522 pp vs. 3.226 pp; small time spreads by 1.806 pp vs. 1.613 pp; checking spreads by 4.100 pp vs. 3.988 pp.

- **Branch quantities (Panel B):** deposit growth is lower for high-beta branches (0.205 vs. 0.262, measured as growth from mid-2003 to mid-2007).
- **Banks (Panel C):** high-beta banks have lower core deposit growth (0.195 vs. 0.298) and lower real estate loan growth (0.253 vs. 0.360), both scaled by initial core deposits.
- **Counties (Panel D):** high-beta counties have lower bank portfolio lending growth (0.098 vs. 0.155) and higher increases in PLS share (0.123 vs. 0.105) and nonbank share (0.032 vs. 0.005).

The same panel shows that high- and low-beta counties differ in market size and structure (e.g., total lending and initial PLS/nonbank shares), motivating later controls.

6.3 Fig. 2: County beta is geographically heterogeneous

Fig. 2 maps CountyBeta_c . The message is that exposure is spatially dispersed, generating identifying cross-sectional variation.



6.4 Fig. 3 and Table 2 (Panel A): the “first stage” ($\beta \rightarrow \text{spread changes}$)

Table 2 (Panel A) estimates Eq. (4) separately for each deposit product:

- Savings: $\gamma = 1.801^{***}$ (s.e. 0.211),
- Small time: $\gamma = 1.056^{***}$ (0.151),
- Interest checking: $\gamma = 0.819^{***}$ (0.080).

Interpretation: pre-boom branch beta strongly predicts how much deposit spreads rise during the 2003–2006 tightening episode. Fig. 3 visualizes these positive relationships with binned scatters.

6.5 Fig. 4 and Table 2 (Panel B): deposit growth falls with beta (including within-bank)

Table 2 (Panel B) estimates Eq. (5) for branch-level deposit growth (June 2003 to June 2007):

- Without bank fixed effects: $\gamma = -0.321^{***} (0.052)$,
- With bank fixed effects: $\gamma = -0.210^{***} (0.060)$.

Interpretation: deposits grow significantly less at high-beta branches. The effect remains negative and significant even *within the same bank* (with bank fixed effects), supporting a deposit-supply channel rather than purely local loan-demand differences.

Table 2

Deposits channel at the branch-level.

This table examines the deposits channel at the branch level. Panel A presents regressions of the branch-level change in the deposit spread from January 2003 to December 2006 on the branch beta. The branch beta is the equal-weighted average of branch-product betas estimated for the three main deposit products (savings, small time, and interest checking) for all branches in a county using quarterly data before 2002. The outcome variables are the change in the deposit spread for savings deposits (column (1)), small time deposits (column (2)), and interest checking deposits (column (3)). Panel B presents regressions of the branch-level growth in deposits from June 2003 to June 2007 on the branch beta. Bank FE denotes whether the regression includes controls for bank fixed effects. The standard errors are clustered at the county level. ***, **, and * denote significance levels at 1%, 5%, and 10% respectively.

	Savings (1)	Time (2)	Checking (3)
Panel A: Deposit spreads			
Branch beta	1.801*** (0.211)	1.056*** (0.151)	0.819*** (0.080)
Observations	5,194	5,462	5,376
R ²	0.019	0.015	0.027
Panel B: Deposit growth			
Branch beta	- 0.321*** (0.052)	- 0.210*** (0.060)	
Bank FE	No	Yes	
Observations	58,546	56,347	
R ²	0.004	0.186	

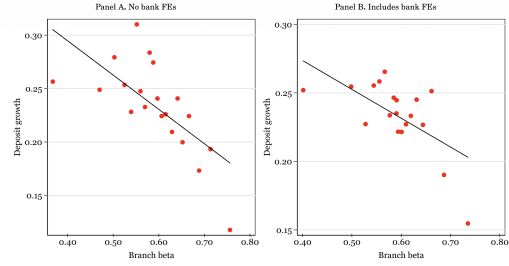


Fig. 4. Branch-level deposit growth. This figure shows binned scatter plots of deposit growth against the branch beta. The branch beta is the same as in Fig. 3. Branches are sorted into 20 bins based on their branch beta. Panel A plots the average deposit growth against the average branch beta in each bin. Panel B plots the same relationship plotted in Panel A after controlling for bank fixed effects.

6.6 Fig. 5: bank-level spread changes and core deposit growth

Fig. 5 repeats the logic at the bank level: higher BankBeta_j is associated with larger increases in the interest-expense spread and lower core deposit growth.

6.7 Table 3 and Fig. 6: bank balance-sheet transmission (deposits → lending)

Panel A (reduced form). Regressing balance-sheet growth (2003–2006, scaled by 2003 core deposits) on bank beta, high-beta banks exhibit:

- lower core deposit growth: $\gamma = -0.385^{***} (0.052)$,
- lower real estate loan growth: $\gamma = -0.364^{***} (0.054)$,
- lower C&I loan growth: $\gamma = -0.084^{***} (0.014)$,
- lower cash and Fed funds repos: $\gamma = -0.078^{***} (0.014)$,
- lower securities: $\gamma = -0.114^{***} (0.023)$,
- lower large time deposits: $\gamma = -0.270^{***} (0.025)$,

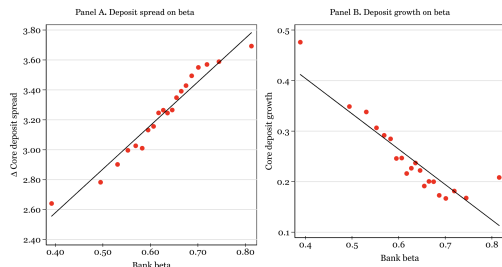


Fig. 5. Bank-level deposit spreads and deposit growth.

This figure shows binned scatter plots of deposit spreads and deposit growth against bank beta. The bank beta is estimated separately for each bank and captures the sensitivity of the deposit spread to changes in the nominal interest rate using quarterly call report data from 1986 to 2002. Banks are sorted into 20 bins based on their bank beta. The figure plots averages by bin. Panel A shows the change in the core deposit spread from January 2003 to December 2006 against the bank beta. Panel B shows core deposit growth from January 2003 to December 2006 against the bank beta. Core deposit growth is measured in dollar differences from 2003 to 2006 and scaled by 2003 core deposits. Bank betas are winsorized at the 1% level; core deposit spread and core deposit growth are both winsorized at the 5% level.

Table 3

Deposits channel at the bank level.

This table presents regressions of the growth in assets (loans, securities, and cash) and liabilities (deposits and equity) on bank beta and core deposits growth. Bank beta is the average spread beta estimated over the years 1986 to 2002. The two panels correspond to two different estimation techniques: reduced-form estimation and two-stage least squares (2SLS). Panel A includes the reduced-form estimates of regressions of growth variables for core deposits (column (1)), real estate (RE) loans (column (2)), commercial and industrial (C&I) loans (column (3)), personal loans (column (4)), cash and Fed funds repos (column (5)), securities (column (6)), large time deposits (i.e., time deposits greater than \$100,000; column (7)), and wholesale deposits (column (8)) on bank beta. Panel B presents the second-stage estimates from 2SLS regressions (corresponding to the first-stage regressions on bank beta in Panel A) for these all of these variables (except core deposits growth) on core deposits growth. The growth variables are calculated as dollar-difference changes from 2003 to 2006 divided by 2003 core deposits. Bank betas are winsorized at the 1% level, and the growth variables are each winsorized at the 5% level. All regressions include controls for the natural logarithm of total bank assets, the equity ratio, and the loan-to-assets ratio as of December 2002. Standard errors are robust. ***, **, and * denote significance levels at 1%, 5%, and 10% respectively.

	Assets								Liabilities	
	Δ Core dep. Core dep. (1)	Δ RE loans Core dep. (2)	Δ C&I loans Core dep. (3)	Δ Personal loans Core dep. (4)	Δ Cash Core dep. (5)	Δ Securities Core dep. (6)	Δ Large time Core dep. (7)	Δ Wholesale Core dep. (8)		
Panel A: Reduced form estimation										
Bank beta	-0.385*** (0.052)	-0.364*** (0.054)	-0.084*** (0.014)	-0.007 (0.005)	-0.078*** (0.014)	-0.114*** (0.023)	-0.270*** (0.025)	-0.019 (0.014)		
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes		
Observations	6,441	6,431	6,441	6,431	6,441	6,441	6,431	6,441		
R ²	0.104	0.164	0.037	0.004	0.026	0.023	0.109	0.049		
Panel B: 2SLS estimation										
Core dep. Core dep.		0.944*** (0.099)	0.218*** (0.032)	0.019 (0.013)	0.202*** (0.037)	0.295*** (0.057)	0.700*** (0.086)	0.050 (0.036)		
Controls		Yes	Yes	Yes	Yes	Yes	Yes	Yes		
Observations		6,431	6,441	6,431	6,441	6,441	6,431	6,441		

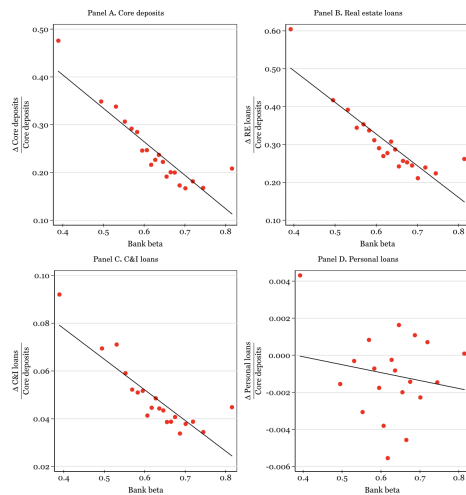


Fig. 6. Bank-level assets and liabilities growth.

This figure shows binned scatter plots of bank core deposits (Panel A), real estate (RE) loans (Panel B), commercial and industrial (C&I) loans (Panel C), personal loans (Panel D), cash and Fed funds repos (Panel E), securities (Panel F), large time deposits (i.e., time deposits greater than \$100,000; Panel G), and wholesale deposits (Panel H) against bank beta. Bank beta is the average spread beta estimated over the years 1986 to 2002. Banks are sorted into 20 bins based on their bank beta. The figure plots the average growth by bin. The growth variables are calculated as dollar-difference changes from 2003 to 2006 divided by 2003 core deposits. Bank betas are winsorized at the 1% level while the growth variables are each winsorized at the 5% level.

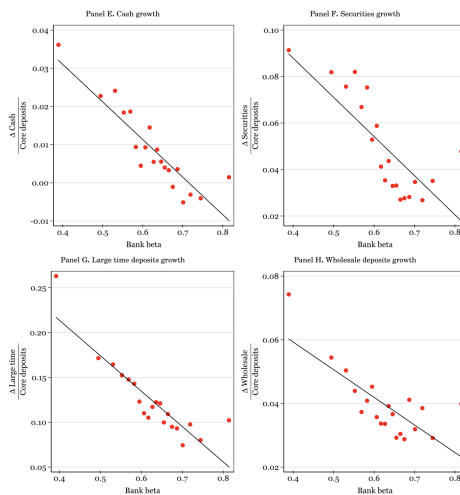


Fig. 6. Continued

- and an economically small, statistically insignificant change in wholesale deposits: $\gamma = -0.019$ (0.014).

Panel B (2SLS pass-through). Using bank beta as an instrument for core deposit growth, the deposit-induced pass-through of a unit increase in core deposits is:

- real estate loans: $\lambda = 0.944^{***}$ (0.099),
- C&I loans: $\lambda = 0.218^{***}$ (0.032),
- cash and Fed funds repos: $\lambda = 0.202^{***}$ (0.037),
- securities: $\lambda = 0.295^{***}$ (0.057),
- large time deposits: $\lambda = 0.700^{***}$ (0.086),
- and no statistically significant increase in wholesale deposits: $\lambda = 0.050$ (0.036).

Interpretation: the deposits instrument primarily moves *real estate lending*, implying a strong bank balance-sheet channel from core deposits to mortgage-related asset growth; Fig. 6 visualizes these linear relationships via binned scatters.

Table 4

Bank portfolio lending.

This table presents regressions of the growth in bank portfolio lending from 2003 to 2006 on county beta. County beta is the average beta of all banks lending in the county, weighted by 2002 lending shares. Column (2) adds controls for the size of the mortgage market (total lending, employment, and median income). Column (3) adds controls for the structure of the mortgage market measured as of 2002 (bank portfolio lending, PLS lending share, nonbank lending share, county-weighted deposit beta, and top four lenders share). The county deposit-weighted beta is the average beta of all banks that raise deposits in a county, weighted by 2002 deposit shares. The top 4 lenders' share is the combined share of the top four lenders in 2002. Column (4) adds controls for loan demand (Bank GSE lending growth, nonbank GSE lending growth, employment growth, and income growth). Standard errors are robust. ***, **, and * denote significance levels at 1%, 5%, and 10% respectively.

	Δ Bank portfolio lending			
	(1)	(2)	(3)	(4)
County beta	-0.748*** (0.164)	-0.827*** (0.159)	-0.522*** (0.177)	-0.596*** (0.177)
Log total lending (2002)		-0.134*** (0.014)	0.094 (0.094)	0.101 (0.092)
Log employment (2002)		0.073*** (0.016)	0.088*** (0.016)	0.102*** (0.016)
Log median income (2002)		0.165*** (0.046)	0.065 (0.046)	0.008 (0.045)
Log bank portfolio lending (2002)			-0.263*** (0.095)	-0.269*** (0.093)
PLS lending share (2002)			0.134 (0.215)	0.084 (0.215)
Nonbank lending share (2002)			0.036 (0.083)	-0.019 (0.082)
County deposit-weighted beta (2002)			0.177 (0.149)	0.352** (0.146)
Top 4 lenders' share (2002)			-0.285*** (0.077)	-0.416*** (0.076)
Δ Bank GSE lending				0.120*** (0.017)
Δ Nonbank GSE lending				0.038*** (0.012)
Δ Employment				0.427*** (0.111)
Δ Income				0.149 (0.186)
Constant	0.533*** (0.090)	-1.137** (0.475)	-0.438 (0.481)	0.038 (0.472)
Observations	3,000	2,999	2,999	2,751
R ²	0.007	0.073	0.139	0.177

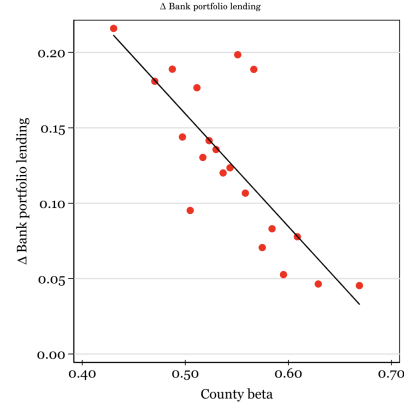


Fig. 7. Bank portfolio lending
This figure presents a binned scatter plot of the growth in bank portfolio lending from 2003 to 2006 against the county beta. The county beta is the average beta of all banks lending in the county, weighted by 2002 mortgage lending shares. Counties are sorted into 20 bins based on their county beta. The figure plots the average growth in bank portfolio lending for the counties in each bin.

6.8 Table 4 and Fig. 7: bank portfolio lending contracts in high-beta counties

Table 4 estimates Eq. (7) with Δ bank portfolio lending (2003–2006) as the outcome. The county-beta coefficient is negative and statistically significant in all specifications:

$$\hat{\gamma}_1 = -0.748^{***}, \quad \hat{\gamma}_2 = -0.827^{***}, \quad \hat{\gamma}_3 = -0.522^{***}, \quad \hat{\gamma}_4 = -0.596^{***},$$

so that $\hat{\gamma} \in [-0.827, -0.522]$ across columns.

Using the reported percentiles of county beta (95th: 0.642; 5th: 0.459), the univariate estimate (column (1)) implies that portfolio lending growth is

$$(-0.748) \times (0.642 - 0.459) = (-0.748) \times 0.183 \approx -0.137,$$

i.e., about **13.7% lower** in a high-beta county (95th percentile) relative to a low-beta county (5th percentile). Fig. 7 visualizes this negative relationship using a binned scatter plot.

6.9 Table 5 and Fig. 8: PLS share rises in high-beta counties

Table 5 uses the change in PLS share as the outcome. The key coefficient is positive:

$$\gamma \approx 0.190 \text{ to } 0.204 \quad (\text{highly significant}).$$

Using the same beta percentiles, the paper reports that the PLS share rises by about

$$0.190 \times (0.642 - 0.459) \approx 0.035,$$

i.e., **3.5 percentage points more** in high-beta counties. Fig. 8 shows the same pattern.

Table 5

PLS lending share.

This table presents regressions of the change in the PLS lending share from 2003 to 2006 on the county beta. PLS lending share is the change in market share of PLS from 2003 to 2006. The county beta is the average beta of all banks lending in the county, weighted by their 2002 lending shares. The control variables in columns (2) to (4) are the same as those in Table 4. Standard errors are robust. ***, **, and * denote significance levels at 1%, 5%, and 10% respectively.

	Δ PLS lending share			
	(1)	(2)	(3)	(4)
County beta	0.190*** (0.047)	0.205*** (0.047)	0.141*** (0.051)	0.204*** (0.047)
Log total lending (2002)		0.003 (0.004)	- 0.042 (0.027)	- 0.035 (0.024)
Log employment (2002)		0.006 (0.005)	0.009* (0.005)	0.013*** (0.004)
Log median income (2002)		- 0.026* (0.013)	0.015 (0.013)	0.025** (0.012)
Log bank portfolio lending (2002)			0.042 (0.027)	0.027 (0.025)
PLS lending share (2002)			- 0.206*** (0.061)	- 0.280*** (0.056)
Nonbank lending share (2002)			0.009 (0.024)	0.049** (0.022)
County deposit-weighted beta (2002)			- 0.094** (0.043)	- 0.167*** (0.038)
Top 4 lenders' share (2002)			- 0.067*** (0.022)	- 0.002 (0.020)
Δ Bank GSE lending				- 0.025*** (0.005)
Δ Nonbank GSE lending				- 0.002 (0.003)
Δ Employment				0.117*** (0.029)
Δ Income				- 0.074 (0.049)
Constant	0.011 (0.025)	0.206 (0.139)	0.028 (0.138)	- 0.093 (0.124)
Observations	3,028	3,027	3,027	2,755
R ²	0.005	0.012	0.120	0.188

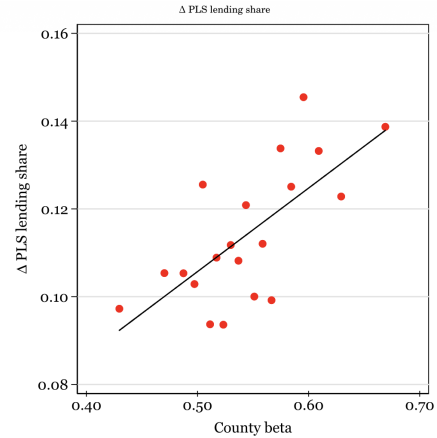


Fig. 8. PLS lending share.
This figure presents a binned scatter plot of the change in the PLS lending share from 2003 to 2006 against the county beta. The county beta is the average beta of all banks lending in the county, weighted by 2002 mortgage lending shares. Counties are sorted into 20 bins based on their county beta. The figure plots the average change in the PLS lending share for the counties in each bin.

6.10 Table 6 and Fig. 9: total bank lending falls, but less than portfolio lending

Table 6 studies total bank lending (portfolio plus bank PLS). The fully controlled estimate is:

$$\gamma \approx -0.372 \quad (0.100).$$

Interpretation: banks partly substitute toward PLS, so total bank lending contracts less than pure portfolio lending, but it still falls in high-beta counties.

Table 6

Total bank lending.

This table presents regressions of the total bank lending growth from 2003 to 2006 on the county beta. Total bank lending is the sum of bank portfolio lending and bank PLS lending. The county beta is the average beta of all banks lending in the county, weighted by their 2002 lending shares. The control variables in columns (2) to (4) are the same as those in Table 4. Standard errors are robust. ***, **, and * denote significance levels at 1%, 5%, and 10% respectively.

	Δ Bank lending			
	(1)	(2)	(3)	(4)
County beta	- 0.863*** (0.137)	- 0.925*** (0.133)	- 0.412*** (0.145)	- 0.372*** (0.144)
Log total lending (2002)		- 0.139*** (0.012)	- 0.146* (0.077)	- 0.139* (0.075)
Log employment (2002)		0.102*** (0.013)	0.123*** (0.013)	0.147*** (0.013)
Log median income (2002)		0.156*** (0.038)	0.160*** (0.038)	0.093*** (0.036)
Log bank portfolio lending (2002)			- 0.044 (0.077)	- 0.066 (0.075)
PLS lending share (2002)			0.108 (0.175)	0.005 (0.174)
Nonbank lending share (2002)			0.665*** (0.068)	0.689*** (0.066)
County deposit-weighted beta (2002)			- 0.053 (0.122)	- 0.024 (0.118)
Top 4 lenders' share (2002)			- 0.276*** (0.063)	- 0.308*** (0.062)
Δ Bank GSE lending				0.073*** (0.014)
Δ Nonbank GSE lending				0.016* (0.010)
Δ Employment				0.695*** (0.090)
Δ Income				0.164 (0.151)
Constant	0.874*** (0.075)	- 0.953** (0.398)	- 1.300*** (0.395)	- 0.769** (0.382)
Observations	3,020	3,019	3,019	2,754
R ²	0.013	0.081	0.177	0.238

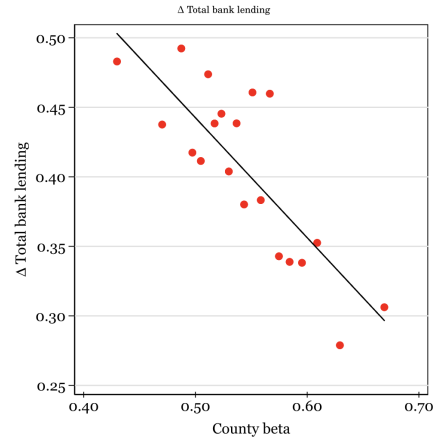


Fig. 9. Total bank lending.
This figure presents a binned scatter plot of the growth in total bank lending from 2003 to 2006 against the county beta. The county beta is the average beta of all banks lending in the county, weighted by 2002 mortgage lending shares. Counties are sorted into 20 bins based on their county beta. The figure plots the average growth in nonbank lending for the counties in each bin.

6.11 Table 7 and Fig. 10: nonbanks gain share in high-beta counties

Table 7 studies the change in the nonbank lending share. The fully controlled estimate is:

$$\gamma \approx 0.169 \quad (0.043).$$

Using the beta percentile gap, this corresponds to roughly $0.169 \times 0.183 \approx 0.031$, i.e., about **3.1 percentage points** more nonbank share growth in high-beta counties. Fig. 10 visualizes the positive relationship.

6.12 Table 8: total non-GSE lending (net quantity effect)

Table 8 aggregates non-GSE lending across bank and nonbank channels. The raw relation is negative, but with full controls the estimate becomes small and insignificant:

$$\gamma \approx -0.117 \quad (0.124).$$

Table 7

Nonbank lending share.

This table presents regressions of the change in the nonbank lending share from 2003 to 2006 on the county beta. Nonbank lending share is the change in market share of nonbanks from 2003 to 2006. The county beta is the average beta of all banks lending in the county, weighted by their 2002 lending shares. The control variables in columns (2) to (4) are the same as those in Table 4. Standard errors are robust. ***, **, and * denote significance levels at 1%, 5%, and 10% respectively.

	Δ Nonbank lending share			
	(1)	(2)	(3)	(4)
County beta	0.219*** (0.041)	0.219*** (0.041)	0.112** (0.045)	0.169*** (0.043)
Log total lending (2002)		0.014*** (0.004)	0.020 (0.024)	0.015 (0.022)
Log employment (2002)		- 0.009** (0.004)	- 0.010** (0.004)	- 0.006 (0.004)
Log median income (2002)		0.019 (0.012)	- 0.003 (0.012)	0.000 (0.011)
Log bank portfolio lending (2002)			- 0.002 (0.024)	- 0.002 (0.023)
PLS lending share (2002)			0.046 (0.054)	0.008 (0.052)
Nonbank lending share (2002)			- 0.323*** (0.021)	- 0.300*** (0.020)
County deposit-weighted beta (2002)			- 0.014 (0.038)	- 0.044 (0.036)
Top 4 lenders' share (2002)			- 0.054*** (0.019)	- 0.042** (0.019)
Δ Bank GSE lending				- 0.021*** (0.004)
Δ Nonbank GSE lending				0.004 (0.003)
Δ Employment				0.070*** (0.027)
Δ Income				- 0.015 (0.045)
Constant	- 0.101*** (0.023)	- 0.289** (0.123)	0.091 (0.123)	0.035 (0.115)
Observations	3,028	3,027	3,027	2,755
R ²	0.009	0.027	0.124	0.161

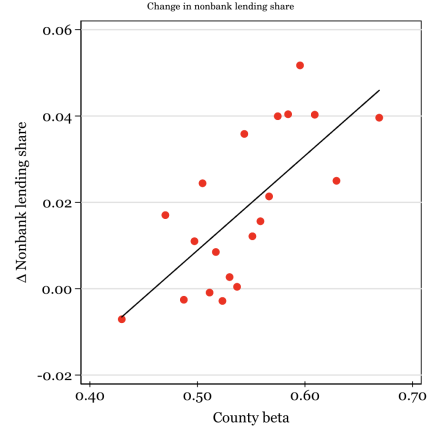


Fig. 10. Nonbank lending share.
This figure presents a binned scatter plot of the change in the nonbank lending share from 2003 to 2006 against the county beta. The county beta is the average beta of all banks lending in the county, weighted by 2002 mortgage lending shares. Counties are sorted into 20 bins based on their county beta. The figure plots the average change in the nonbank lending share for the counties in each bin.

Interpretation: tightening changes composition more than total quantity.

6.13 Table 9: GSE lending

Table 9 shows that total GSE lending growth is not robustly related to county beta. **Interpretation:** the deposits channel is primarily about the non-GSE/PLS-vs-portfolio margin, not the GSE channel.

6.14 Table 10: including refinancing

Table 10 repeats the main results including refinancing.

- Portfolio lending (incl. refi): $\gamma \approx -0.703$ (0.138),
- PLS share (incl. refi): $\gamma \approx 0.165$ (0.039).

Interpretation: the mechanism is not specific to purchase mortgages.

6.15 Table 11: controlling for risky-lending proxies

Adding HMDA denial rates and a subprime-lender share does not remove the county-beta effects. In the richest specification:

- portfolio lending: $\gamma \approx -0.450$ (0.174),
- PLS share: $\gamma \approx 0.160$ (0.045).

Interpretation: the beta effect is not simply a proxy for local “riskiness” trends.

Table 8

Total lending.

This table presents regressions of the total non-GSE lending growth from 2003 to 2006 on the county beta. Total non-GSE lending is the sum of bank portfolio lending, bank and nonbank PLS lending, and bank lending. The county beta is the average beta of all banks lending in the county, weighted by their 2002 lending shares. The control variables in columns (2) to (4) are the same as those in Table 4. Standard errors are robust. ***, **, and * denote significance levels at 1%, 5%, and 10% respectively.

	Δ Total lending			
	(1)	(2)	(3)	(4)
County beta	-0.477*** (0.118)	-0.547*** (0.114)	-0.208 (0.128)	-0.117 (0.124)
Log total lending (2002)		-0.127*** (0.010)	-0.166*** (0.068)	-0.170*** (0.064)
Log employment (2002)		0.095*** (0.012)	0.115*** (0.012)	0.137*** (0.011)
Log median income (2002)		0.210*** (0.033)	0.186*** (0.033)	0.095*** (0.033)
Log bank portfolio lending (2002)		-0.006 (0.068)	-0.017 (0.065)	
PLS lending share (2002)		0.223 (0.154)	0.140 (0.149)	
Nonbank lending share (2002)		0.182*** (0.059)	0.209*** (0.057)	
County deposit-weighted beta (2002)		-0.075 (0.107)	-0.080 (0.101)	
Top 4 lenders' share (2002)		-0.373*** (0.055)	-0.392*** (0.053)	
Δ Bank GSE lending		0.039*** (0.012)	0.095*** (0.012)	
Δ Nonbank GSE lending		0.022*** (0.008)	0.022*** (0.008)	
Δ Employment		0.813*** (0.077)		
Δ Income		0.196 (0.065)		
Constant	0.687*** (0.065)	-1.709*** (0.341)	-1.519*** (0.347)	-0.751** (0.329)
Observations	3,028	3,027	2,755	2,755
R ²	0.005	0.075	0.126	0.186

Table 9

GSE lending.

This table presents regressions of the total GSE lending growth from 2003 to 2006 on the county beta. The county beta is the average beta of all banks lending in the county, weighted by their 2002 lending shares. The control variables in Columns (2) to (4) are the same as those in Table 4 (except for Bank GSE and Nonbank GSE lending growth in column (4)). Standard errors are robust. ***, **, and * denote significance levels at 1%, 5%, and 10% respectively.

	Δ Total GSE lending			
	(1)	(2)	(3)	(4)
County beta	-0.199 (0.150)	-0.213 (0.142)	0.091 (0.163)	0.126 (0.162)
Log total lending (2002)		-0.068*** (0.012)	0.047 (0.086)	0.048 (0.085)
Log employment (2002)		-0.001 (0.014)	-0.005 (0.015)	0.016 (0.015)
Log median income (2002)		-0.194*** (0.040)	-0.211*** (0.042)	-0.240*** (0.042)
Log bank portfolio lending (2002)		-0.115 (0.086)	-0.134 (0.086)	
PLS lending share (2002)		-0.024 (0.196)	-0.073 (0.195)	
Nonbank lending share (2002)		-0.059 (0.075)	-0.088 (0.075)	
County deposit-weighted beta (2002)		-0.469*** (0.136)	-0.406*** (0.135)	
Top 4 lenders' share (2002)		0.075 (0.070)	0.028 (0.070)	
Δ Employment		0.676*** (0.105)		
Δ Income		-0.556*** (0.178)		
Constant	0.009 (0.082)	2.400*** (0.420)	2.634*** (0.438)	3.062*** (0.439)
Observations	2,992	2,991	2,991	2,991
R ²	0.001	0.118	0.128	0.142

Table 10

Home purchases and refinancing.

This table presents regressions that include refinancings when computing lending growth. Panel A presents the same specifications as in Table 4 using the growth in the sum of bank portfolio lending and refinancings as the outcome variable. Panel B presents the same specifications as in Table 5 using the change in the PLS market share as the outcome variable. The PLS share is computed from the sum of purchase loans and refinancings. The coefficients on the control variables are not shown. ***, **, and * denote significance levels at 1%, 5%, and 10% respectively.

Panel A: Bank portfolio lending				
	Δ Bank portfolio lending			
	(1)	(2)	(3)	(4)
County beta	-0.732*** (0.133)	-0.725*** (0.126)	-0.620*** (0.140)	-0.703*** (0.138)
Characteristics controls		Yes	Yes	Yes
Market-structure controls		Yes	Yes	Yes
Δ Demand controls				
Observations	3,031	3,030	3,030	2,922
R ²	0.010	0.117	0.166	0.205
Panel B: PLS lending share				
	Δ PLS lending share			
	(1)	(2)	(3)	(4)
County beta	0.139*** (0.037)	0.161*** (0.037)	0.150*** (0.040)	0.185*** (0.039)
Characteristics controls		Yes	Yes	Yes
Market-structure controls		Yes	Yes	Yes
Δ Demand controls				
Observations	3,033	3,032	3,032	2,923
R ²	0.005	0.043	0.148	0.166

6.16 Table 12: pre-trends

Controlling for pre-2002 trends (and, where available, house-price trends) leaves the results intact. With the richest pre-trend controls:

- portfolio lending: $\gamma \approx -0.758$ (0.218),
- PLS share: $\gamma \approx 0.169$ (0.056).

6.17 Table 13: weighted regressions

Weighting by population, market size, or deposits yields similar qualitative conclusions. For example, deposit-weighted WLS gives:

- portfolio lending: $\gamma \approx -0.780$ (0.123),
- PLS share: $\gamma \approx 0.178$ (0.030).

Table 11

Risky-loan analysis.

This table presents regressions that control for the risky-loan share. Column (1) of Panel A presents the benchmark specification from column (4) in Table 4. Columns (2) adds controls for the HMDA denial rate. The HMDA denial rate is computed as the number of denials divided by the sum of denials, originations, and "approved not accepted." Column (3) adds controls for the subprime lending share. Subprime lender share is computed as the number of non-GSE loans originated by subprime lenders divided by the total number of non-GSE originations. The list of subprime lenders is from the United States Department of Housing and Urban Development (HUD). Column (4) adds controls for both the HMDA denial rate and the subprime lending share. Panel B presents the same specifications using PLS market share as the outcome variable. The coefficients on the other control variables are not shown. ***, **, and * denote significance levels at 1%, 5%, and 10% respectively.

Panel A: Bank portfolio lending				
	Δ Bank portfolio lending			
	(1)	(2)	(3)	(4)
County beta	-0.596*** (0.177)	-0.613*** (0.174)	-0.493*** (0.176)	-0.450*** (0.174)
Δ HMDA denial rate		-1.854*** (0.148)	-1.893*** (0.146)	
HMDA denial rate (2002)		-0.628*** (0.119)	-0.387*** (0.121)	
Δ Subprime lender share		-1.390*** (0.123)	-1.254*** (0.122)	
Subprime lender share (2002)		-0.685*** (0.138)	-0.526*** (0.138)	
Controls	Yes	Yes	Yes	Yes
Observations	2,751	2,727	2,751	2,727
R ²	0.177	0.224	0.215	0.253
Panel B: PLS lending share				
	Δ PLS lending share			
	(1)	(2)	(3)	(4)
County beta	0.204*** (0.047)	0.199*** (0.046)	0.170*** (0.046)	0.169*** (0.045)
Δ HMDA denial rate		0.197*** (0.039)	0.141*** (0.038)	
HMDA denial rate (2002)		-0.063*** (0.032)	-0.158*** (0.031)	
Δ Subprime lender share		0.409*** (0.032)	0.429*** (0.031)	
Subprime lender share (2002)		0.197*** (0.036)	0.249*** (0.036)	
Controls	Yes	Yes	Yes	Yes
Observations	2,755	2,730	2,755	2,730
R ²	0.188	0.210	0.236	0.264

Table 12

Pre-trend analysis.

This table presents regressions that control for pre-trends in lending growth, housing prices, and median household income. Column (1) of Panel A presents the benchmark specification from column (4) in Table 4. Column (2) adds controls for the growth in total lending from 1996 to 2002, the change in the PLS lending share from 1996 to 2002, and the growth in median household income from 1996 to 2002. Column (3) restricts the sample to counties with Zillow housing price data (1,582 counties). Column (4) adds the controls from column (2) as well as controls for the growth in the Zillow house price from 1996 to 2002. Panel B presents the same specifications as in Table 5 using PLS market share as the outcome variable. The coefficients on the other control variables are not shown. ***, **, and * denote significance levels at 1%, 5%, and 10% respectively.

Panel A: Bank portfolio lending				
	Δ Bank portfolio lending			
	(1)	(2)	(3)	(4)
County beta	-0.596*** (0.177)	-0.662*** (0.180)	-0.739*** (0.214)	-0.758*** (0.218)
Δ Total lending (96-02)		-0.002 (0.016)	-0.013 (0.020)	
Δ PLS lending share (96-02)		0.125*** (0.046)	0.047 (0.055)	
Δ Income (96-02)		0.012 (0.147)	0.086 (0.159)	
Δ Home value (96-02)		-0.024 (0.046)	-0.024 (0.046)	
Controls	Yes	Yes	Yes	Yes
Observations	2,751	2,750	1,582	1,582
R ²	0.177	0.179	0.268	0.269
Panel B: PLS lending share				
	Δ PLS lending share			
	(1)	(2)	(3)	(4)
County beta	0.204*** (0.047)	0.225*** (0.047)	0.153*** (0.055)	0.169*** (0.056)
Δ Total lending (96-02)		0.012*** (0.004)	0.005 (0.005)	
Δ PLS lending share (96-02)		-0.019 (0.012)	-0.022 (0.014)	
Δ Income (96-02)		0.025 (0.039)	0.005 (0.041)	
Δ Home value (96-02)		0.010 (0.012)	0.010 (0.012)	
Controls	Yes	Yes	Yes	Yes
Observations	2,755	2,754	1,582	1,582
R ²	0.188	0.192	0.297	0.299

Table 13

Weighted least squares.

This table presents regressions weighted by population, loan market size, and deposits. Panel A presents the benchmark specification from column (4) in Table 4 using each of the following variables as weights: population (column (1)), bank portfolio lending (column (2)), total lending (column (3)), and deposits (column (4)) as of 2002. Panel B presents the same specifications using PLS market share as the outcome variable. The coefficients on the other control variables are not shown. ***, **, and * denote significance levels at 1%, 5%, and 10% respectively.

Panel A: Bank portfolio lending				
	Δ Bank portfolio lending			
	Population weights (1)	Bank portfolio lending weights (2)	Total lending weights (3)	Deposits weights (4)
County beta	-0.588*** (0.133)	-0.398*** (0.131)	-0.530*** (0.127)	-0.780*** (0.123)
Controls	Yes	Yes	Yes	Yes
Observations	2,750	2,751	2,751	2,751
R ²	0.424	0.459	0.544	0.443
Panel B: PLS lending share				
	Δ PLS lending share			
	Population weights (1)	Bank portfolio lending weights (2)	Total lending weights (3)	Deposits weights (4)
County beta	0.206*** (0.032)	0.207*** (0.032)	0.222*** (0.030)	0.178*** (0.030)
Controls	Yes	Yes	Yes	Yes
Observations	2,754	2,755	2,755	2,755
R ²	0.368	0.434	0.456	0.395

6.18 Table 14: 2014–2017 tightening episode

Using realized betas for 2014–2017, the paper finds:

- portfolio lending: $\gamma \approx -0.675$ (0.136),
- PLS share: $\gamma \approx 0.117$ (0.038),
- total bank lending: $\gamma \approx -0.323$ (0.115),
- total lending: $\gamma \approx -0.176$ (0.096; marginal significance).

Interpretation: the composition channel remains relevant in a modern tightening cycle.

6.19 Fig. 11: aggregate counterfactual

Fig. 11 converts the cross-sectional estimates into an illustrative aggregate counterfactual: remove the impact of tightening through the deposits channel. The key implication is that total non-GSE lending would have been only modestly higher, but **PLS would have been much lower and portfolio lending much higher**. This underscores the paper’s central message: *monetary policy affects the composition and fragility of credit supply, not only its level*.

Table 14
2014–2017 analysis.
This table presents regressions of the growth in bank portfolio lending, PLS lending share, bank lending, nonbank lending share, and total lending from 2014 to 2017 on the county beta. The county beta is the average realized beta of all banks lending in the county, weighted by their 2013 lending shares. Realized betas are calculated as the difference between year-end 2017 and 2014 interest expense rates divided by the difference in year-end 2017 and 2014 Fed funds rate. Realized betas are winsorized at the 1% level. All models include as controls the 2013 levels of log total lending, log employment, log median income, log bank portfolio lending, PLS lending share, nonbank lending share, top 4 lenders’ share, and realized county deposit-weighted beta. Further controls include the 2014–2017 county-level growth in bank GSE lending, nonbank GSE lending, employment, and income. The coefficients on the control variables are omitted for compactness. “***”, “**”, and “*” denote significance levels at 1%, 5%, and 10% respectively.

	Δ Bank portfolio lending (1)	Δ PLS lending share (2)	Δ Total bank lending (3)	Δ Nonbank lending share (4)	Δ Total lending (5)
County beta	-0.675*** (0.136)	0.117*** (0.038)	-0.323*** (0.115)	0.062 (0.039)	-0.176* (0.096)
Controls	Yes	Yes	Yes	Yes	Yes
Observations	2,727	2,746	2,742	2,746	2,746
R ²	0.095	0.045	0.074	0.048	0.116

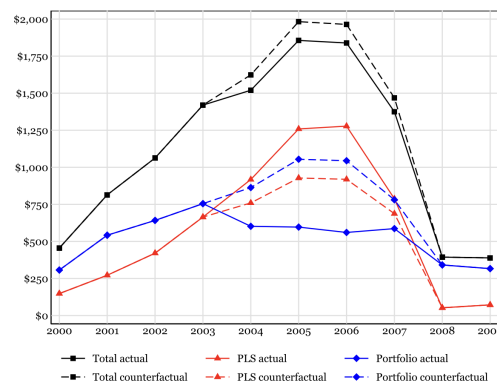


Fig. 11. Actual and estimated counterfactual originations.
The figure shows actual (solid lines) and estimated counterfactual (dashed lines) aggregate originations of PLS (red triangles), bank portfolio loans (blue squares), and total non-GSE loans (black diamonds) from 2000 to 2009. The counterfactual originations remove the impact of Fed tightening through the deposits channel. This impact is based on our cross-sectional estimates; hence, it does not account for potential spillovers. Actual PLS originations are from the Securities Industry and Financial Markets Association (SIFMA) and actual portfolio loans are from the Home Mortgage Disclosure Act (HMDA) dataset. Total loans are the sum of PLS and portfolio loans. See Section 5 for details and discussion. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

7 From cross section to aggregate: the counterfactual mapping (how Fig. 11 is built)

The aggregation is intentionally transparent:

1. Use Table 8 to infer the counterfactual change in aggregate total non-GSE lending absent tightening through the deposits channel.
2. Use Table 5 to infer the counterfactual PLS share path (the paper’s baseline mapping keeps the PLS share roughly constant absent tightening).

3. Convert counterfactual shares and totals into counterfactual dollar volumes for PLS and portfolio lending.

The paper emphasizes that this mapping is *partial equilibrium*: it does not model spillovers, feedback, or other channels of monetary policy.

8 Identification assumptions and limitations

- **Loan demand confounds:** the branch-level within-bank design is a key argument that deposit results are not driven by local mortgage demand. County-level results rely on rich controls and pre-trend tests.
- **What betas capture:** betas are exposure measures (how spreads respond to policy), reflecting local competition and depositor composition. The paper does not separately identify each structural source.
- **Aggregation requires assumptions:** mapping cross-sectional effects into aggregate counterfactuals necessarily abstracts from general equilibrium and endogenous PLS capacity. The paper discusses these issues explicitly.

9 Conclusion

9.1 Summary

The paper resolves a central puzzle from the 2003–2006 housing boom: *the Federal Reserve tightened aggressively (the Fed funds rate rose by 4.25 percentage points), yet mortgage lending and the housing boom accelerated*. The paper concludes that monetary policy *did matter*, but primarily by **changing the composition and funding fragility of mortgage credit**, not by mechanically shrinking total mortgage lending.

The full logic is one continuous chain:

1. **Identification via the deposits channel.** The paper uses cross-sectional heterogeneity in banks’ exposure to the *deposits channel of monetary policy*, measured by the *deposit-spread beta*: banks with higher betas expand their deposit spreads more when the Fed tightens (they keep deposit rates relatively low when market rates rise), which triggers larger deposit outflows and a larger contraction in deposit funding.
2. **Fed tightening reduced insured deposit funding where exposure was high.** During 2003–2006, counties served by high-beta banks experienced significantly weaker deposit growth, consistent with a tightening-induced contraction in deposit supply.
3. **Banks responded by contracting portfolio mortgage lending.** The contraction in deposits translated into a large contraction in *bank portfolio* mortgage lending (the paper summarizes the implied effect as a **32%** contraction in portfolio mortgage lending due to tightening, relative to a counterfactual without tightening through this channel).

4. **But total mortgage lending did not contract much because lending migrated to private-label securitization (PLS).** The bank contraction was largely offset by substitution toward **privately securitized (PLS) mortgages**, predominantly originated by **nonbank** lenders. In other words, tighter monetary policy shifted mortgage intermediation away from banks rather than shutting it down.
5. **The key macro implication is a compositional shift toward fragile funding.** Bank portfolio mortgages are funded by government-insured deposits (stable, run-resistant), whereas PLS mortgages rely on capital markets funding and lack an explicit government backstop. Therefore, even if aggregate lending is not greatly reduced, tightening induced a shift *from stable to run-prone funding*, increasing mortgage-market fragility and exposure to wholesale funding runs and market freezes (as in 2007–2008).
6. **Aggregate counterfactual: small quantity effect, large composition effect.** Mapping the cross-sectional estimates to an aggregate counterfactual, the paper concludes that total non-GSE lending in 2006 would have been only modestly higher absent tightening through the deposits channel (the paper reports an implied gap of roughly **6.4%**). However, the *composition* would have been dramatically different. For example, the paper reports that actual PLS rose from about \$665B (2003) to \$1,278B (2006), while bank portfolio lending fell from about \$755B to \$561B. Under a counterfactual in which tightening did not induce a shift to PLS, PLS in 2006 would have been much lower (about \$919B) and portfolio lending much higher (about \$1,045B), implying that most of the boom would have been intermediated inside the banking system rather than through PLS.
7. **Policy interpretation: monetary policy alone is an imperfect tool for controlling mortgage credit when substitution is easy.** The findings speak directly to the debate between the view that tighter policy could have prevented the boom and the view that tighter supervision of lending standards was needed. The paper’s evidence implies that tightening can reduce bank credit supply, but capital markets intermediation can substitute, so restraining overall mortgage credit (and especially risky origination) likely requires complementary tools (e.g., tighter lending standards and supervision).
8. **External validity: similar dynamics reappear in the 2014–2017 tightening cycle.** When the Fed tightened again after 2014 and PLS lending reemerged, deposit growth slowed, banks pulled back on portfolio lending, and PLS expanded to fill part of the gap, consistent with the same mechanism operating outside the original boom period.

9.2 Economic intuition: why tightening can accelerate a shift to fragile mortgage finance

The core intuition is that monetary tightening hits the mortgage market *through bank funding*, but the economy can re-route mortgage credit through nonbank capital markets.

1) Tightening makes deposit funding scarcer where banks have deposit market power.

When the Fed raises rates, depositors have more attractive outside options (e.g., money-market instruments). Banks with high deposit market power do not raise deposit rates one-for-one, so their deposit spreads rise and deposits flow out. Hence, tightening reduces insured deposit funding disproportionately at high-beta banks.

2) Mortgage portfolio lending is the margin that depends on insured deposits.

Banks can fund some assets in wholesale markets, but portfolio mortgages are naturally tied to stable deposit funding and balance-sheet capacity. When deposits contract, banks reduce balance-sheet-intensive lending (portfolio mortgages), especially in segments not easily offloaded.

3) PLS is the substitution margin that is not deposit-funded.

Private-label securitization allows mortgages to be funded by investors in capital markets rather than by insured deposits. Nonbank originators can scale origination by relying on securitization pipelines and warehouse funding, so when banks retreat, they can expand and substitute into the gap.

4) Quantity versus composition.

Because the supply curve for mortgages is an envelope across multiple intermediaries, tightening may have a limited effect on *total* mortgage quantity if nonbanks can substitute. But the *composition* changes sharply, shifting credit toward a funding structure that is more sensitive to runs and freezes.

5) Why this matters for crises.

A boom financed with insured deposits is more insulated from sudden funding runs. A boom financed with PLS and wholesale funding is more vulnerable to a sharp reversal in investor risk appetite, which is precisely the fragility observed in 2007–2008.

9.3 What to remember

- The paper’s key contribution is to separate the **bank-credit-supply effect** of tightening (via deposits) from the **system-wide mortgage-credit outcome** once **nonbank substitution** is allowed.
- Monetary policy can have a small effect on aggregate mortgage lending but a large effect on the **shadow-bank share** and therefore on **financial fragility**.
- A clean empirical template is: use **predetermined exposure measures** (deposit-spread betas) to identify heterogeneous responses to aggregate policy changes in the cross section.

9.4 Takeaway

Fed tightening during 2003–2006 substantially reduced banks’ insured-deposit funding and portfolio mortgage lending, but mortgage credit largely migrated to private-label securitization led by nonbanks, so monetary policy reshaped the mortgage market by shifting it from stable deposit-funded intermediation toward fragile capital-markets-funded intermediation, limiting its ability to curb total lending while increasing run risk.